Homework 07 - Analysis of Variance

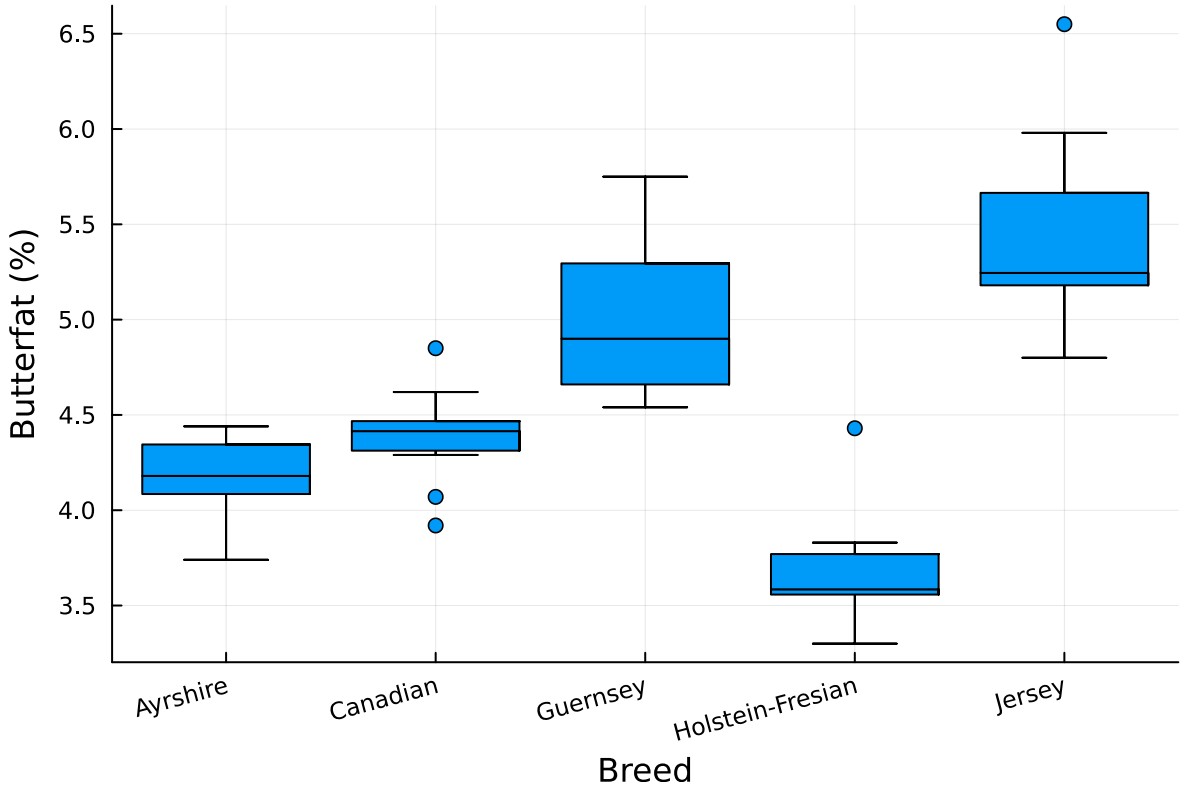
```
using RData, DataFrames, CategoricalArrays, GLM, Statistics, StatsPlots, HypothesisTests
, Distributions
```

	Butterfat	Breed	Age
1	3.74	"Ayrshire"	"Mature"
2	3.77	"Ayrshire"	"Mature"
3	4.1	"Ayrshire"	"Mature"
4	4.27	"Ayrshire"	"Mature"
5	4.11	"Ayrshire"	"Mature"
6	4.44	"Ayrshire"	"Mature"

```
begin
    # butterfat data set from the faraway R package (CRAN)
    tgz = download("https://cran.r-
    project.org/src/contrib/Archive/faraway/faraway_1.0.8.tar.gz")
    dir = mktempdir()
    run(`tar xzf $tgz -C $dir faraway/data/butterfat.rda`)
    butterfat = load(joinpath(dir, "faraway/data/butterfat.rda"))["butterfat"]
    data1 = subset(butterfat, :Age => a -> a .== "Mature")
    first(data1, 6)
end
```

Problem 1

Part a)




```
boxplot(string.(data1.Breed), data1.Butterfat, legend=false, xlabel="Breed",
ylabel="Butterfat (%)", xrotation=15)
```

Jersey and Guernsey cows have the highest butterfat content, Holstein-Fresian the lowest, and Ayrshire and Canadian are in between and look similar to each other. The spread also differs by breed: Jersey and Guernsey are more variable (with a few high outliers) than Holstein-Fresian, Ayrshire, and Canadian.

Part b)

stats =

	Breed	Mean	SD
1	"Ayrshire"	4.154	0.24627
2	"Canadian"	4.389	0.260532
3	"Guernsey"	5.001	0.408315
4	"Holstein-Fresian"	3.676	0.310133
5	"Jersey"	5.452	0.506377

```
stats = combine(groupby(data1, :Breed), :Butterfat => mean => :Mean, :Butterfat => std
=> :SD)
```

The largest standard deviation (Jersey, ≈ 0.51) is about twice the smallest (Ayrshire, ≈ 0.25). This is below the usual 2:1 rule-of-thumb cutoff (ratio ≈ 2.06, borderline), so the constant variance assumption is not badly violated, but it is worth checking in the diagnostics. The variance also tends to increase with the mean.

Part c)

Effect coding (sum-to-zero contrasts), with Jersey as the level coded –1 so that Breed1 = Ayrshire, Breed2 = Canadian, Breed3 = Guernsey, Breed4 = Holstein-Fresian (same as R's `contr.sum`).

```
model =
StatsModels.TableRegressionModel{LinearModel{GLM.LmResp{Vector{Float64}}, GLM.DensePredChol{
```

```
Butterfat ~ 1 + Breed

Coefficients:
```

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	4.5344	0.0509057	89.07	<1e-51	4.43187	4.63693
Breed: Ayrshire	-0.3804	0.101811	-3.74	0.0005	-0.585459	-0.175341
Breed: Canadian	-0.1454	0.101811	-1.43	0.1602	-0.350459	0.0596585
Breed: Guernsey	0.4666	0.101811	4.58	<1e-04	0.261541	0.671659
Breed: Holstein-Fresian	-0.8584	0.101811	-8.43	<1e-10	-1.06346	-0.653341

```
model = lm(@formula(Butterfat ~ Breed), data1; contrasts = Dict(:Breed =>
EffectsCoding(base = "Jersey")))
```

β = [4.5344, -0.3804, -0.1454, 0.4666, -0.8584]

```
β = coef(model)
```

	Breed	Computation	Mean
1	"Ayrshire"	"β₀ + β₁"	4.154
2	"Canadian"	"β₀ + β₂"	4.389
3	"Guernsey"	"β₀ + β₃"	5.001
4	"Holstein-Fresian"	"β₀ + β₄"	3.676
5	"Jersey"	"β₀ − β₁ − β₂ − β₃ − β₄"	5.452

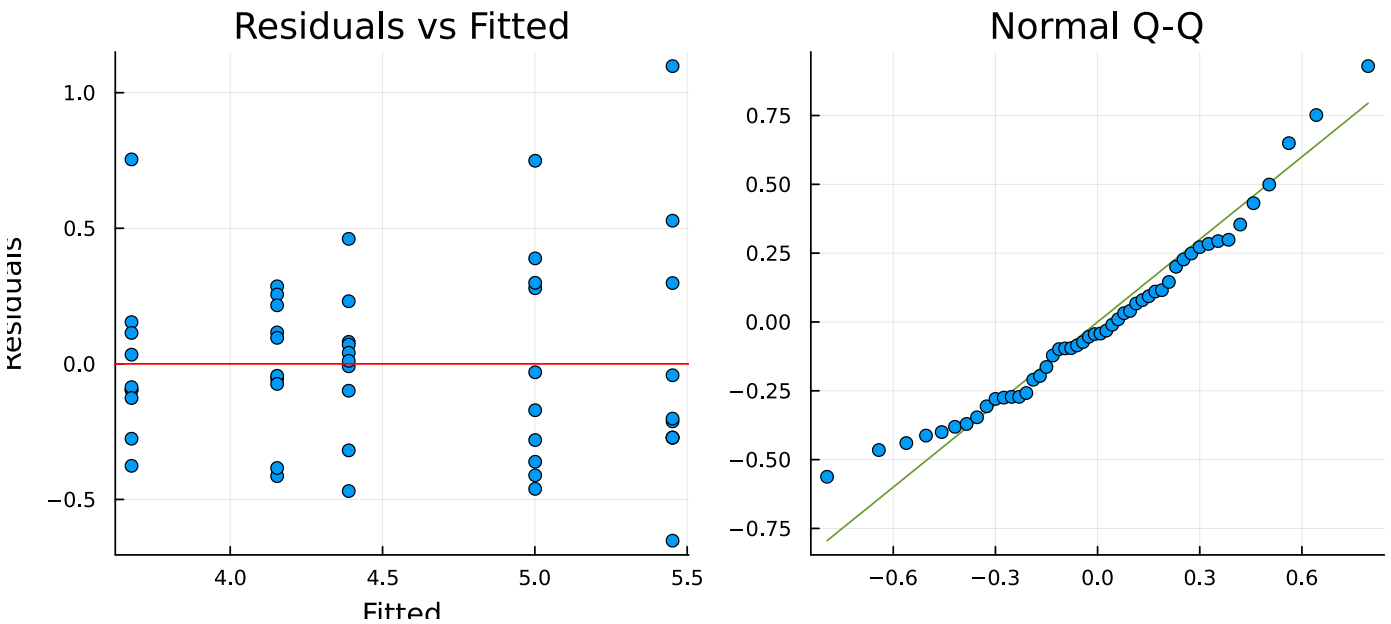
```
DataFrame(Breed = levels(data1.Breed),
    Computation = ["β₀ + β₁", "β₀ + β₂", "β₀ + β₃", "β₀ + β₄", "β₀ − β₁ − β₂ − β₃ − β₄"],
    Mean = [β[1] + β[2], β[1] + β[3], β[1] + β[4], β[1] + β[5], β[1] - sum(β[2:5])])
```

Butterfat = 4.5344 − 0.3804·Breed1 − 0.1454·Breed2 + 0.4666·Breed3 − 0.8584·Breed4

- Ayrshire: 4.5344 − 0.3804 = 4.154
- Canadian: 4.5344 − 0.1454 = 4.389
- Guernsey: 4.5344 + 0.4666 = 5.001
- Holstein-Fresian: 4.5344 − 0.8584 = 3.676
- Jersey: 4.5344 − (−0.3804 − 0.1454 + 0.4666 − 0.8584) = 4.5344 + 0.9176 = 5.452

These match the means in Part b. The intercept (4.5344) is the grand mean, the average of the five breed means; each coefficient is how far that breed's mean is from the grand mean.

Part d)




```
begin
    p1 = scatter(fitted(model), residuals(model), xlabel="Fitted", ylabel="Residuals",
    title="Residuals vs Fitted", legend=false)
    hline!(p1, [0], color=:red)
    p2 = qqnorm(residuals(model), title="Normal Q-Q", legend=false)
    plot(p1, p2, layout=(1, 2), size=(800, 350))
end
```

```
Shapiro-Wilk normality test
---------------------------
Population details:
    parameter of interest:   Squared correlation of data and expected order statistics of N(
    value under h_0:         1.0
    point estimate:          0.955489

Test summary:
    outcome with 95% confidence: fail to reject h_0
    one-sided p-value:           0.0576

Details:
    number of observations: 50
    censored ratio:         0.0
    W-statistic:            0.955489
```

```
ShapiroWilkTest(residuals(model))
```

```
Levene's test
-------------
Population details:
    parameter of interest:   Variances
    value under h_0:         "all equal"
    point estimate:          NaN

Test summary:
    outcome with 95% confidence: fail to reject h_0
    p-value:                     0.1135

Details:
    number of observations: [10, 10, 10, 10, 10]
    W statistic:            1.98171
    degrees of freedom:     (4, 45)
```

```
LeveneTest([data1.Butterfat[data1.Breed .== b] for b in levels(data1.Breed)]...)
```

- **Constant variance:** the residuals vs fitted plot shows somewhat more spread at the higher fitted values (Guernsey and Jersey), but Levene's test does not reject equal variances ($p \approx 0.11$).
- **Normality:** the Q-Q plot is mostly linear with a slightly heavy upper tail from a few large residuals; Shapiro-Wilk does not reject normality at $\alpha = 0.05$ ($p \approx 0.058$).
- **Independence:** the cows are different animals, so independence is reasonable.

The assumptions are acceptable, though the variance and right tail are borderline.

Part e)

```
F-test: 2 models fitted on 50 observations
──────────────────────────────────────────────────────────────────
     DOF  ΔDOF      SSR      ΔSSR      R²      ΔR²       F*   p(>F)
──────────────────────────────────────────────────────────────────
[1]    2        25.4546            0.0000
[2]    6     4   5.8306  -19.6240  0.7709  0.7709  37.8639  <1e-13
──────────────────────────────────────────────────────────────────
```

```
ftest(lm(@formula(Butterfat ~ 1), data1).model, model.model)
```

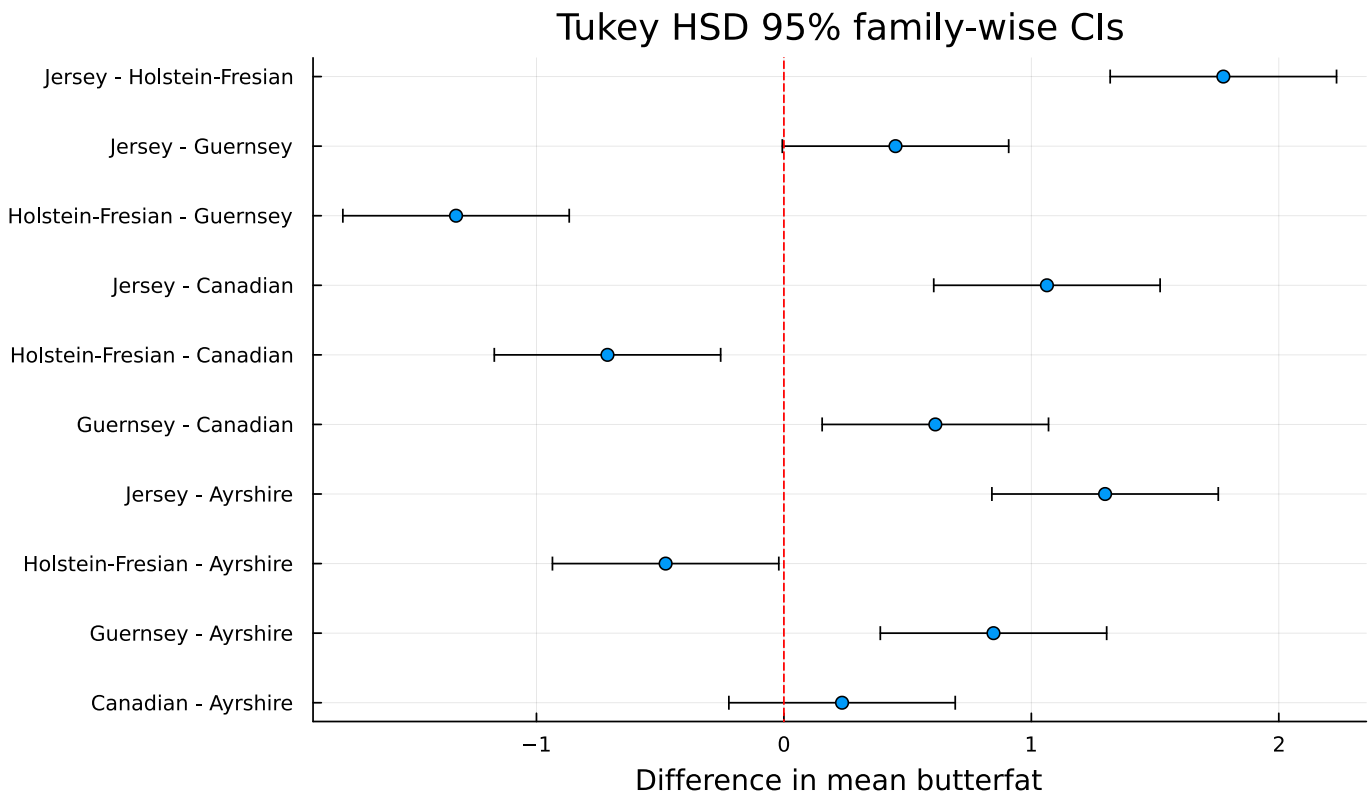
- H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (equivalently $\beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$). Mean butterfat is the same for all breeds.
- H_a: at least one breed mean is different.
- $\alpha = 0.05$
- Test statistic: $F = 37.86$ on 4 and 45 df
- p-value < 0.0001
- Decision: $p < \alpha$, so reject H_0.
- Conclusion: there is strong evidence that mean butterfat content differs between at least two of the breeds of mature cows.

Part f)

Tukey HSD pairwise comparisons (95% family-wise confidence intervals)

	Comparison	Diff	Lower	Upper	p
1	"Canadian - Ayrshire"	0.235	-0.22241	0.69241	0.593206
2	"Guernsey - Ayrshire"	0.847	0.38959	1.30441	3.6557e-5
3	"Holstein-Fresian - Ayrshire"	-0.478	-0.93541	-0.0205895	0.0365566
4	"Jersey - Ayrshire"	1.298	0.84059	1.75541	2.75927e-9
5	"Guernsey - Canadian"	0.612	0.15459	1.06941	0.00375194
6	"Holstein-Fresian - Canadian"	-0.713	-1.17041	-0.25559	0.000550156
7	"Jersey - Canadian"	1.063	0.60559	1.52041	3.87254e-7
8	"Holstein-Fresian - Guernsey"	-1.325	-1.78241	-0.86759	1.57755e-9
9	"Jersey - Guernsey"	0.451	-0.00641049	0.90841	0.0549908
10	"Jersey - Holstein-Fresian"	1.776	1.31859	2.23341	4.83613e-13

```
begin
    lv = levels(data1.Breed)
    μ = [mean(data1.Butterfat[data1.Breed .== b]) for b in lv]
    n = 10
    dfe = dof_residual(model)
    se = sqrt(deviance(model) / dfe / n)
    q = quantile(StudentizedRange(dfe, length(lv)), 0.95)
    tukey = DataFrame(Comparison = String[], Diff = Float64[], Lower = Float64[], Upper
    = Float64[], p = Float64[])
    for i in 1:length(lv), j in i+1:length(lv)
        d = μ[j] - μ[i]
        push!(tukey, ("$(lv[j]) - $(lv[i])", d, d - q * se, d + q * se,
        ccdf(StudentizedRange(dfe, length(lv)), abs(d) / se)))
    end
    tukey
end
```




```
begin
    scatter(tukey.Diff, 1:nrow(tukey), xerror=(tukey.Diff .- tukey.Lower, tukey.Upper
    .- tukey.Diff), yticks=(1:nrow(tukey), tukey.Comparison), legend=false,
    xlabel="Difference in mean butterfat", title="Tukey HSD 95% family-wise CIs", size=
    (800, 450))
    vline!([0], color=:red, linestyle=:dash)
end
```

The intervals that contain 0 are **Canadian – Ayrshire** ($p \approx 0.59$) and **Jersey – Guernsey** ($p \approx 0.055$). These two pairwise differences are not statistically significant. All the other pairs are significantly different at the 0.05 level.